ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2022-2023**

DURATION: 3 HOURS **FULL MARKS: 150**

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) What are the steps of problem-solving in engineering? With a suitable example, briefly explain how numerical methods are used to solve an engineering problem. $2+1$ (CO2) (PO1)

Solution:

The steps of problem-solving in engineering are —

1. Describing or formulating the problem.
2. Using a mathematical model to represent the problem.
3. Solving the mathematical model.
4. Using the solution of the mathematical model to solve the original problem.

e.g. Let's consider a kinematics problem. Given the initial velocity u and the height h of a falling body, we can calculate the time t it needs to reach the ground by finding the positive root of the equation $ut + \frac{1}{2}gt^2 - h = 0$ using a root-finding algorithm.

b) i. Suppose, R and k are arbitrary numbers such that $R > 0$ and $k \neq 0$. Prove that the Newton-Raphson formula for finding the kth root of R, which is mathematically denoted as $\sqrt[k]{R}$, is as shown in Equation-1. 5 (CO2) (PO1)

$$x_{i+1} = \frac{1}{k}\left[(k-1)x_i + \frac{R}{x_i^{k-1}}\right] \tag{1}$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} & x = \sqrt[k]{R} \\ \Rightarrow \quad & x^k = R \\ \Rightarrow \quad & x^k - R = 0 \end{aligned}$$

We take $f(x) = x^k - R$ and so $f'(x) = kx^{k-1}$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^k - R}{kx_i^{k-1}} \\ &= \frac{kx_i^k - x_i^k + R}{kx_i^{k-1}} \\ &= \frac{(k-1)x_i^k + R}{kx_i^{k-1}} \\ &= \frac{1}{k}\left[\frac{(k-1)x_i^k + R}{x_i^{k-1}}\right] \\ &= \frac{1}{k}\left[(k-1)x_i + \frac{R}{x_i^{k-1}}\right] \quad \square \text{ Q.E.D.} \end{aligned}$$

ii. Draw a rough sketch of the graph of two functions $f_1(x)$ and $f_2(x)$ given that — $2+2$ (CO3) (PO2)

- For $f_1(x)$, False Position method performs better than the Bisection method.
- For $f_2(x)$, Bisection method performs better than the False Position method.

Here, *"performs better"* means that it takes a lesser number of iterations to converge to the root. You have the liberty to choose the initial guesses $\langle x_l, x_u \rangle$ as well.

Solution:

$f_1(x)$ may look something like,

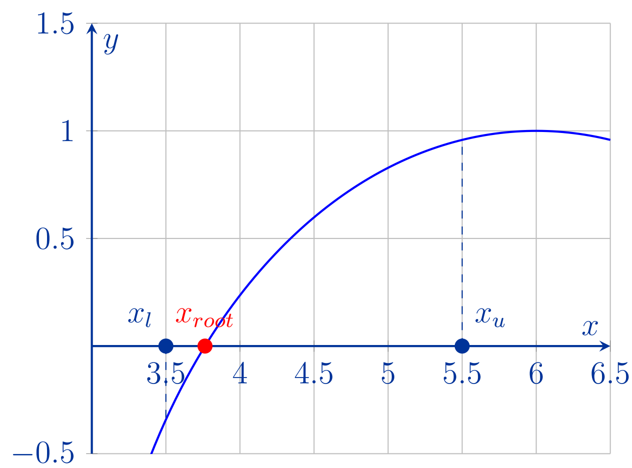

Figure 1: Graph of $f_1(x)$.

$f_2(x)$ may look something like,

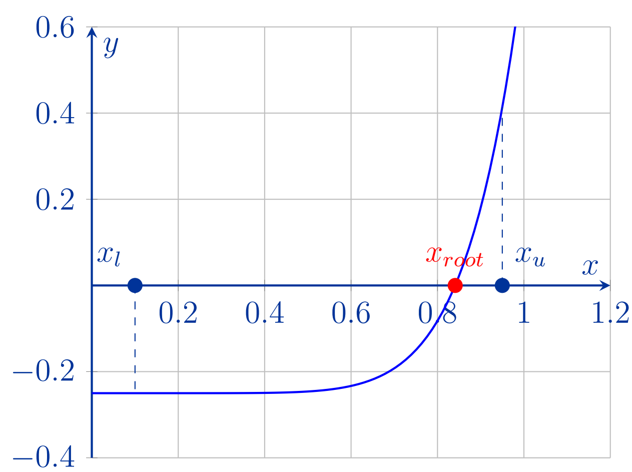

Figure 2: Graph of $f_2(x)$.

iii. Derive the formula for the Secant method geometrically. 4 (CO4) (PO1)

Solution:

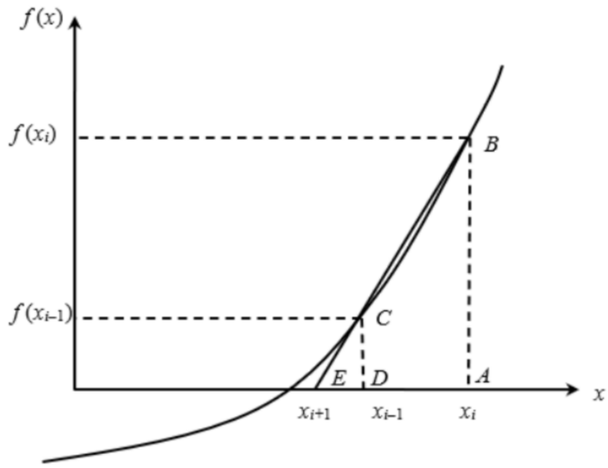

Figure 3: Geometrical representation of the Secant method.

Here, ABE and DCE are similar triangles.
Hence,

$$\frac{AB}{AE} = \frac{DC}{DE}$$

$$\frac{f(x_i)}{x_i - x_{i+1}} = \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}}$$

$$\Rightarrow f(x_i)(x_{i-1} - x_{i+1}) = f(x_{i-1})(x_i - x_{i+1})$$

$$\Rightarrow f(x_i)x_{i-1} - x_{i+1}f(x_i) = x_i f(x_{i-1}) - x_{i+1}f(x_{i-1})$$

$$\Rightarrow x_{i+1}f(x_{i-1}) - x_{i+1}f(x_i) = f(x_{i-1})x_i - f(x_i)x_{i-1}$$

$$\Rightarrow x_{i+1} = \frac{x_i f(x_{i-1}) - f(x_i)x_{i-1}}{f(x_{i-1}) - f(x_i)} + x_i - x_i$$

$$\Rightarrow x_{i+1} = x_i + \frac{x_i f(x_{i-1}) - f(x_i)x_{i-1} - x_i f(x_{i-1}) + x_i f(x_i)}{f(x_{i-1}) - f(x_i)}$$

$$\Rightarrow x_{i+1} = x_i + \frac{-f(x_i)(-x_{i-1} + x_i)}{f(x_{i-1}) - f(x_i)}$$

$$\Rightarrow x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Hence, we get the formula for the Secant method,

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

c) The *Bronze Ratio*, often denoted by the Greek letter β (beta), is the ratio of two quantities a and b, such that $a > b > 0$, if it satisfies the equality $\dfrac{3a+b}{a} = \dfrac{a}{b}$. The constant β is equal to the positive root of a particular function $f(x)$. The numerical value of that root is,

$$x^{+}_{root} = \beta = \frac{3+\sqrt{13}}{2} = 3.302775638\ldots$$

i. From the given scenario, formulate the function $f(x)$. 3 (CO1) (PO2)

Solution:
According to the given condition,

$$\frac{3a+b}{a} = \frac{a}{b} = \beta$$

$$\Rightarrow \quad \frac{3a}{a} + \frac{b}{a} - \frac{a}{b} = 0 \quad \Rightarrow \quad 3 + \frac{1}{\beta} - \beta = 0 \quad \Rightarrow \quad 3\beta + 1 - \beta^2 = 0$$

$$\Rightarrow \quad \beta^2 - 3\beta - 1 = 0$$

$$\Rightarrow \quad x^2 - 3x - 1 = 0$$

So, the function is $f(x) = x^2 - 3x - 1$.

ii. Apply the Secant method to estimate the value of the bronze ratio β. The initial guesses are $x_{-1} = 1.5$ and $x_0 = 2$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 on your answer script and fill it out after performing the necessary calculations. 11 (CO2) (PO1)

Table 1: The relevant values obtained in the answer of Question 1.c)

Iteration	x_{i-1}	x_i	x_{i+1}	$\lvert\epsilon_a\rvert\%$	m	$f(x_{i+1})$
1	1.5	2				
2						
3						
4						

Solution:
After performing the series of calculations, we end up with the following table.

Iteration	x_{i-1}	x_i	x_{i+1}	$\lvert\epsilon_a\rvert\%$	m	$f(x_{i+1})$
1	1.5	2	8	75	0	39
2	2	8	2.428	229.41	0	-2.387
3	8	2.428	2.75	11.69	0	-1.687
4	2.428	2.75	3.524	21.98	0	0.849

2. a) "The magnitude of the true error and the magnitude of the approximate error do not necessarily reflect how bad the error is." — Do you agree? Justify your answer. 4 (CO3) (PO1)

Solution:

Yes, I agree with the given statement.
Let's consider an arbitrary function $f_1(x)$. Suppose we get a true error value of $E_t = p$, where p is a small number, while computing the value of $f_1'(2)$ with $h = 0.3$. Now, if we try to do the same thing for another function $f_2(x)$, where $f_2(x) = 10^{-6} \times f_1(x)$, we will get an even more minuscule true error value of $E_t = p \times 10^{-6}$. Even though the problems are the same in essence, we are getting a smaller true error value in the 2nd case, which is giving us an improper interpretability or a false impression of the goodness of the approximation. The same argument holds true for the approximate error as well.

b) The Taylor series for a function $f(x)$ is — 15 (CO2) (PO1)

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + f'''''(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \text{cosec}(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.
We know, $\frac{d}{dx}(\text{cosec}(x)) = -\text{cosec}(x)\cot(x)$, and $\text{cosec}(0)$ = undefined. So we will get undefined values for $f(0)$ and $f^i(0)$.
Instead let's consider, $f(x) = \text{cosec}(x) = \dfrac{1}{\sin(x)}$, and find the Maclaurin series of the denominator $g(x) = \sin(x)$. Let's calculate its derivatives at the point $x = 0$.

$$\begin{aligned}
g(x) &= \sin(x), \quad g(0) = 0 \\
g'(x) &= \cos(x), \quad g'(0) = 1 \\
g''(x) &= -\sin(x), \quad g''(0) = 0 \\
g'''(x) &= -\cos(x), \quad g'''(0) = -1 \\
g''''(x) &= \sin(x), \quad g''''(0) = 0
\end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$g(0+h) = g(0) + g'(0)\frac{h}{1!} + g''(0)\frac{h^2}{2!} + g'''(0)\frac{h^3}{3!} + g''''(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow g(h) = 0 + \frac{h}{1!} + 0 - \frac{h^3}{3!} + 0 + \dots$$

$$\Rightarrow g(x) = \frac{x}{1!} - \frac{x^3}{3!} + \dots \Rightarrow \sin(x) = \frac{x}{1!} - \frac{x^3}{3!} + \dots = x - \frac{x^3}{6} + \dots$$

Now, for the original function $f(x) = \text{cosec}(x)$,

$$\text{cosec}(x) = \frac{1}{x - \frac{x^3}{6} + \dots}$$

Performing algebraic long division, we get,

$$\begin{array}{r|l} & \frac{1}{x} + \frac{x}{6} + \dots \\ \hline x - \frac{x^3}{6} + \dots & 1 + 0 \\ & 1 - \frac{x^2}{6} + \dots \\ \hline & \quad \frac{x^2}{6} + \dots \\ & \quad \frac{x^2}{6} + \dots \\ \hline & \quad 0 + \dots \\ & \quad \vdots \end{array}$$

Hence, we can write,

$$\Rightarrow \operatorname{cosec}(x) = \frac{1}{x} + \frac{x}{6} + \dots \qquad \text{[Quotient of the above algebraic long division.]}$$

c) We know, $\sin\left(\frac{\pi}{2}\right) = 1$ and $\cos\left(\frac{\pi}{2}\right) = 0$. Using this information and the Taylor series (with at least the first 5 terms) approximate the value of $\sin(2)$. 8 (CO2) (PO1)
Note: The unit of angular measure that has been used here is radians (rad).

Solution:
Let, $x = \frac{\pi}{2}$ and $x + h = 2$. This implies, $h = 2 - x = 2 - \frac{\pi}{2} = 0.4292$.
Let's calculate its derivatives at the point $x = \frac{\pi}{2}$.

$$f(x) = \sin(x), \quad f\left(\frac{\pi}{2}\right) = 1$$
$$f'(x) = \cos(x), \quad f'\left(\frac{\pi}{2}\right) = 0$$
$$f''(x) = -\sin(x), \quad f''\left(\frac{\pi}{2}\right) = -1$$
$$f'''(x) = -\cos(x), \quad f'''\left(\frac{\pi}{2}\right) = 0$$
$$f''''(x) = \sin(x), \quad f''''\left(\frac{\pi}{2}\right) = 1$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \dots$$

For $x = \frac{\pi}{2}$,

$$f\left(\frac{\pi}{2} + h\right) = f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\frac{h}{1!} + f''\left(\frac{\pi}{2}\right)\frac{h^2}{2!} + f'''\left(\frac{\pi}{2}\right)\frac{h^3}{3!} + f''''\left(\frac{\pi}{2}\right)\frac{h^4}{4!} + \dots$$

$$\begin{aligned} \Rightarrow f\left(\frac{\pi}{2} + 0.4292\right) &= 1 + 0 - \frac{(0.4292)^2}{2!} + 0 + \frac{(0.4292)^4}{4!} + \dots \\ &= 1 + 0 - 0.092106 + 0 + 0.00141393 + \dots \\ &\approx 0.90931 \end{aligned}$$

3. a) Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 6 (CO4) (PO1)

Solution:

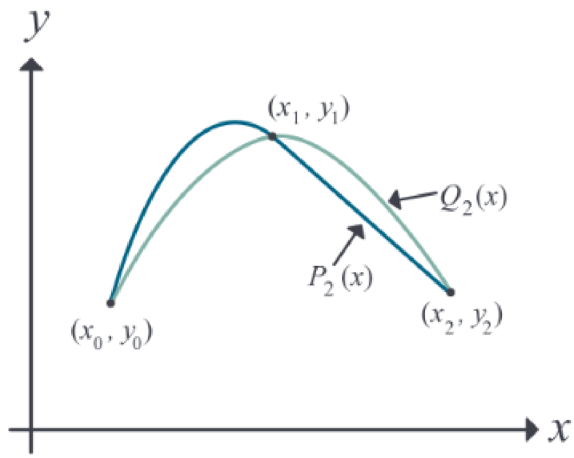

Figure 4: Interpolating polynomials through three points.

Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n+1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n+1$ data points (x_0, y_0), (x_1, y_1), $\ldots$, (x_n, y_n). Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n+1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$

Hence

$$R_n(x_i) = P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \ldots, n\}$$
$$\Rightarrow R_n(x_i) = 0$$

The nth order polynomial $R_n(x)$ has $n+1$ zeros. A polynomial of order n can have $n+1$ zeros only if it is identical to a zero polynomial, that is,

$$R_n(x) \equiv 0$$
$$\Rightarrow P_n(x) \equiv Q_n(x)$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. □ Q.E.D.

b) Compare and contrast the following methods of interpolation — Direct method, Newton's Divided Difference method, and Lagrangian method. 4 (CO3) (PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	**Direct**	**Newton's Divided Difference**	**Lagrangian**
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$	$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1)\ldots(x - x_{n-1})$	$f_n(x) = \sum_{i=0}^{n} \left(\prod_{j=0, j \neq i}^{n} \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(\|A\| + \|b\|))$, or $O((n+1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

c) A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Table 2 portrays multiple recorded observations involving a thermistor. 8 (CO2) (PO1)

Table 2: Temperature T of a thermistor as a function of its resistance R for Question-3.c)

Resistance, R (Ω)	**Temperature,** T ($°C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the Newton's Divided Difference method of interpolation and a second order polynomial.

Solution:

The 2nd order NDD interpolant is —

$$T(R) = b_0 + b_1(R - R_0) + b_2(R - R_0)(R - R_1)$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the coefficients.

$$b_0 = T(R_0) = 30.131$$

$$b_1 = \frac{T(R_1) - T(R_0)}{R_1 - R_0} = \frac{40.12 - 30.131}{636 - 911.3} = -0.0362$$

$$b_2 = \frac{\dfrac{T(R_2) - T(R_1)}{R_2 - R_1} - \dfrac{T(R_1) - T(R_0)}{R_1 - R_0}}{R_2 - R_0} = \frac{\dfrac{50.128 - 40.12}{451.1 - 636} - \dfrac{40.12 - 30.131}{636 - 911.3}}{451.1 - 911.3} = 3.877 \times 10^{-5}$$

Hence,

$$\begin{aligned} T(R) &= b_0 + b_1(R - R_0) + b_2(R - R_0)(R - R_1) \\ &= 30.131 - 0.0362(R - 911.3) + 3.877 \times 10^{-5}(R - 911.3)(R - 636); \quad 451.1 \le R \le 911.3 \end{aligned}$$

Now that we have the NDD interpolant, we can obtain the value of $T(754.8)$.

$$\begin{aligned} T(754.8) &= 30.131 - 0.0362(754.8 - 911.3) + 3.877 \times 10^{-5}(754.8 - 911.3)(754.8 - 636) \\ &\approx 35.075°C \end{aligned}$$

4. a) The general linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ is $6+6$ (CO4) (PO1)

$$y = a_0 + a_1 x$$

where a_0 and a_1 are the two parameters of the regression model.

i. Derive the formulae for finding the optimal values of a_0 and a_1.

Solution:
The residual error for each given data point is,

$$E_i = y_i - a_0 - a_1 x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^{n} E_i^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2$$

To find the optimal a_0 and a_1, we look for the values of a_0 and a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test. Taking the partial derivatives of S_r with respect to a_0 and a_1 and set them to zero.

$$\frac{\partial S_r}{\partial a_0} = 2\sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)(-1) = 0$$

$$\frac{\partial S_r}{\partial a_1} = 2\sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)(-x_i) = 0$$

Dividing both sides by 2 and expanding the summations in the aforementioned equations,

$$-\sum_{i=1}^{n} y_i + \sum_{i=1}^{n} a_0 + \sum_{i=1}^{n} a_1 x_i = 0 \Rightarrow -\sum_i y_i + n a_0 + \sum_i a_1 x_i = 0 \Rightarrow a_0 = \frac{\sum_i y_i}{n} - a_1 \frac{\sum_i x_i}{n}$$

For the 2nd equation, we perform substitution

$$-\sum_{i=1}^{n} y_i x_i + \sum_{i=1}^{n} a_0 x_i + \sum_{i=1}^{n} a_1 x_i^2 = 0 \Rightarrow \sum_i x_i y_i - \left(\frac{\sum_i y_i}{n} - a_1 \frac{\sum_i x_i}{n}\right) \sum_i x_i - a_1 \sum_i x_i^2 = 0$$

$$\Rightarrow \sum_i x_i y_i - \sum_i x_i \frac{\sum_i y_i}{n} + a_1 \frac{(\sum_i x_i)^2}{n} - a_1 \sum_i x_i^2 = 0$$

$$\Rightarrow \left(\frac{\left(\sum_i x_i\right)^2}{n} - \sum_i x_i^2 \right) = \frac{\sum_i x_i \sum_i y_i}{n} - \sum_i x_i y_i$$

$$\Rightarrow a_1 = \frac{n\sum_i x_i y_i - \sum_i x_i \sum_i y_i}{n\sum_i x_i^2 - \left(\sum_i x_i\right)^2} = \frac{\sum_i x_i y_i - n\bar{x}\bar{y}}{\sum_i x_i^2 - n\bar{x}^2}$$

Plugging in this expression for a_1 to the previous equation,

$$a_0 = \frac{\sum_i x_i^2 \sum_i y_i - \sum_i x_i \sum_i x_i y_i}{n\sum_i x_i^2 - \left(\sum_i x_i\right)^2}$$

ii. Prove that the values of a_0 and a_1 obtained using the formulae derived in Question-4.a)(i) correspond to the absolute minimum of the optimization criterion used.

Solution:
Let's invoke the second derivative for this.
From the 1st-order partial derivatives in the last question, we get,

$$\frac{\partial^2 S_r}{\partial a_0^2} = -2\sum_i -1 = 2n > 0; \quad \text{[2nd condition satisfied.]}$$

$$\frac{\partial^2 S_r}{\partial a_1^2} = 2\sum_i x_i^2 > 0; \quad \text{[2nd condition satisfied.]}$$

$$\frac{\partial^2 S_r}{\partial a_0 \partial a_1} = 2\sum_i x_i$$

Now, let's test the 1st condition of the second derivative test.

$$\frac{\partial^2 S_r}{\partial a_0^2}\frac{\partial^2 S_r}{\partial a_1^2} - \left(\frac{\partial^2 S_r}{\partial a_0 \partial a_1}\right)^2 = (2n)\left(2\sum_i x_i^2\right) - \left(2\sum_i x_i\right)^2$$

$$= 4\left[n\sum_i x_i^2 - \left(\sum_i x_i\right)^2\right] > 0; \quad \text{[1st condition satisfied.]}$$

So, the critical point (a_0, a_1) obtained using the formulae corresponds to the absolute minimum value of S_r.

b) Suppose you are given a set of n data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ and you want to regress the data to an mth order Polynomial model, where $m < n$. How would you choose a suitable value of m? 5 (CO3) (PO1)

Solution:
In order to choose a suitable value of m, we need to maintain a Bias-Variance tradeoff. If the value of m is too high, the polynomial model will overfit the given n data points and result in high variance. On the other hand, if the value of m is too small, the polynomial model will underfit the given n data points and result in high bias.

c) Sodium borohydride ($NaBH_4$) is an inorganic compound that can be used as fuel for fuel cells. The Overpotential (η) vs Current (I) data depicted in Table 3 was obtained during a study conducted to assess its electrochemical kinetics. 8 (CO2) (PO1)

Table 3: Electrochemical Kinetics data of $NaBH_4$ for Question-4.c)

Overpotential, η (V)	**Current, I** (A)
-0.29563	0.00226
-0.24346	0.00212
-0.19012	0.00206
-0.18772	0.00202
-0.13407	0.00199
-0.0861	0.00195

The relationship that exists between the overpotential (η) and current (I) can be represented using the Logarithmic model (also known as the Tafel equation) —

$$\eta = a + b \ln I$$

where a is an electrochemical kinetics parameter of borohydride on the electrode and b is the experimental Tafel slope parameter. Determine the value of a and b using the data points provided in Table 3. You are allowed to apply data transformation/linearization.

Solution:

Suppose, $x = \ln I$ and $y = \eta$. Then we can express the Tafel equation as a straight line $y = a + bx$. We can now obtain the value of the parameters a and b using the linear regression formulae,

$$b = \frac{n \sum_i x_i y_i - \sum_i x_i \sum_i y_i}{n \sum_i x_i^2 - \left(\sum_i x_i\right)^2}$$

$$a = \bar{y} - b\bar{x}$$

Using the $\ln I$ and η values in the calculator, we find the following values,
$\sum_i x_i = -37.0979$, $\sum_i y_i = -1.1371$, $\sum_i x_i^2 = 229.3896$, $\sum_i x_i y_i = 7.0116$.
Let's plug these values into the formulae.

$$b = \frac{6 \times 7.0116 - (-37.0979) \times (-1.1371)}{6 \times 229.3896 - (-37.0979)^2} = -1.3601$$

$$a = \frac{-1.1371}{6} - (-1.3601) \times \frac{-37.0979}{6} = -8.5992$$

Plugging these values into the given Logarithmic model equation, we get,

$$\eta = -8.5992 - 1.3601 \times \ln I$$

5. a) i. Using the method of coefficients, derive the formula for the Single-segment Trapezoidal rule. 7 + 6 (CO3) (PO1)

Solution:

Let's consider the integral to be a weighted sum of the functional values at the two limits a and b.

$$\int_a^b f(x)dx \approx c_1 f(a) + c_2 f(b)$$

This weighted sum is exactly equal to the integral of the straight line $f_1(x) = a_0 + a_1 x = 0$.

$$\int_a^b (a_0 + a_1 x)dx = \left[a_0 x + a_1 \frac{x^2}{2}\right]_a^b = a_0(b-a) + a_1\left(\frac{b^2 - a^2}{2}\right)$$

We know that the straight line passes through the points $(a, f(a))$ and $(b, f(b))$. So we can express the functional values as,

$$f(a) = a_0 + a_1 a$$
$$f(b) = a_0 + a_1 b$$

Then we can write the weighted sum as,

$$\begin{aligned} c_1(a_0 + a_1 a) + c_2(a_0 + a_1 b) =& c_1 a_0 + c_1 a_1 a + c_2 a_0 + c_2 a_1 b \\ =& a_0(c_1 + c_2) + a_1(c_1 a + c_2 b) \end{aligned}$$

Equating the coefficients, we get,

$$c_1 + c_2 = b - a$$
$$c_1 a + c_2 b = \frac{b^2 - a^2}{2}$$

Solving them, we get,

$$c_1 = c_2 = \frac{b-a}{2}$$

Therefore,

$$\begin{aligned}\int_a^b f(x)dx &\approx c_1 f(a) + c_2 f(b) \\ &= \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b)\end{aligned}$$

ii. Applying the formula from Question-5.a)(i), derive the formula for the Multi-segment Trapezoidal rule.

Solution:
Taking a step size of $h = \dfrac{b-a}{n}$, we break down the integral I into n integrals in the following manner,

$$\begin{aligned}I &= \int_a^b f(x)dx \\ &= \int_a^{a+h} f(x)dx + \int_{a+h}^{a+2h} f(x)dx + \cdots + \int_{a+(n-2)h}^{a+(n-1)h} f(x)dx + \int_{a+(n-1)h}^{a+nh=b} f(x)dx \\ &= [(a+h) - a]\left[\frac{f(a) + f(a+h)}{2}\right] + \cdots + [b - (a + (n-1)h)]\left[\frac{f(a + (n-1)h) + f(b)}{2}\right] \\ &= h\left[\frac{f(a) + 2f(a+h) + \cdots + 2f(a+(n-1)h) + f(b)}{2}\right] \\ &= \frac{h}{2}\left[f(a) + 2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\} + f(b)\right] \\ &= \frac{b-a}{2n}\left[f(a) + 2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\} + f(b)\right]\end{aligned}$$

b) The velocity v of a body is given by 5 + 4 (CO2) (PO1)

$$v(t) = \begin{cases} 2t; & \text{if } 1 \leq t \leq 5 \\ 5t^2 + 3; & \text{if } 5 < t \leq 14 \end{cases}$$

where the time t is given in seconds (s), and v is given in meters per second $(\mathrm{m\,s^{-1}})$.

i. Using 2-segment Simpson's 1/3 rule, approximate the distance (S) in meters covered by the body from $t = 2\mathrm{s}$ to $t = 9\mathrm{s}$.

Solution:
The multiple-segment equation for Simpson's 1/3 rule is

$$\int_a^b f(x)dx \approx \frac{b-a}{3n}\left[f(x_0) + 4\sum_{\substack{i=1,\\ i=\text{odd}}}^{n-1} f(x_i) + 2\sum_{\substack{i=2,\\ i=\text{even}}}^{n-2} f(x_i) + f(x_n)\right]$$

We have, $a = 2$, $b = 9$, $n = 2$, $h = \dfrac{b-a}{n} = \dfrac{9-2}{2} = 3.5$. We also know that the distance function $S(t)$ is the integration of the velocity function $v(t)$. The distance covered by the body is, therefore,

$$\begin{aligned} S = \int_2^9 v(t)dt &\approx \frac{9-2}{3\times 2}\left[v(t_0) + 4\sum_{\substack{i=1,\\ i=\text{odd}}}^{2-1} v(t_i) + 2\sum_{\substack{i=2,\\ i=\text{even}}}^{2-2} v(t_i) + v(t_n)\right] \\ &= \frac{9-2}{3\times 2} \times [v(t_0) + 4v(t_1) + v(t_2)] \\ &= 1.1667 \times [2\times 2 + 4\times 5\times 5.5^2 + 3 + 5\times 9^2 + 3] \\ &= 1.1667 \times [4 + 4\times 154.25 + 408] \\ &= 1200.5\,\mathrm{m} \end{aligned}$$

ii. Analytically determine the exact value of the covered distance (S_{exact}) and use it to calculate the true error (E_t) and the absolute relative true error $(|\epsilon_t|\%)$.

Solution:
The exact value of the covered distance is,

$$\begin{aligned} S_{exact} &= \int_2^5 2t dt + \int_5^9 (5t^2 + 3)dt \\ &= \left[\frac{2t^2}{2}\right]_2^5 + \left[\frac{5t^3}{3} + 3t\right]_5^9 \\ &= 1039.6667\,\mathrm{m} \end{aligned}$$

The true error is,

$$E_t = \text{True Value} - \text{Approx. Value} = 1039.6667\,\mathrm{m} - 1200.5\,\mathrm{m} = -160.8333\,\mathrm{m}$$

The absolute relative true error is,

$$|\epsilon_t| = \left|\frac{\text{True Value} - \text{Approx. Value}}{\text{True Value}}\right| \times 100\% = 15.469\%$$

c) Define Eigenvector and Eigenvalue. 3 (CO3) (PO1)

Solution:
The vectors $\vec{v}$ of a vector space that do not get knocked off their own spans upon undergoing linear transformation by a matrix A, but are just stretched or squished by a certain factor λ, are called the eigenvectors of the linear transformation done by A.
Mathematically,

$$A\vec{v} = \lambda\vec{v}$$

Here, the corresponding eigenvalue is the factor by which an eigenvector is stretched or squished. If the eigenvalue is negative, the eigenvector's direction is reversed.

6. a) Compare and contrast the following numerical algorithms for solving 1st-order ordinary differential equations — Euler's method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method. 4 (CO3) (PO1)

Solution:
Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	**Euler**	**Runge-Kutta 2nd Order**	**Runge-Kutta 4th Order**
Formula	$y_{i+1} = y_i + f(x_i, y_i)h$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h$, $k_1 = f(x_i, y_i)$, $k_2 = f(x_i + ph, y_i + qk_1h)$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$
Terms of the Taylor series	First 2 terms.	First 3 terms.	First 5 terms.
Order of the Taylor polynomial	1st order — which is why it's also called the Runge-Kutta 1st order method.	2nd order — which is why it's called the Runge-Kutta 2nd order method.	4th order — which is why it's also called the Runge-Kutta 4th order method.
Local Truncation Error	$O(h^2)$	$O(h^3)$	$O(h^5)$
Global Truncation Error	$O(h)$	$O(h^2)$	$O(h^4)$
Variant Approaches	None.	Has 3 different versions – Heun's method, Midpoint method, and Ralston's method	Has 2 different versions – Runge's approach and Kutta's approach.

b) Determine the approximate value of 6 (CO2) (PO1)

$$I = \int_5^8 6x^3 dx$$

using Euler's method. Use a step size of $h = 1.5$.

Solution:
Let's rewrite the given integral as the solution of an ODE –

$$\frac{dy}{dx} = 6x^3; \quad y(5) = 0$$

where $y(8)$ gives us the value of the integral.
The Euler's method formula is,

$$y_{i+1} = y_i + f(x_i, y_i)h$$

In the 1st step, we have $x_0 = 5$, $y_0 = 0$, and given $h = 1.5$.

$$x_1 = x_0 + h = 5 + 1.5 = 6.5$$
$$y_1 = y_0 + f(x_0, y_0)h = 0 + f(5, 0) \times 1.5$$
$$= 0 + (6 \times 5^3) \times 1.5 = 1125 \approx y(6.5)$$

In the 2nd step, we have $x_1 = 6.5$, $y_1 = 1125$, and as before $h = 1.5$.

$$x_2 = x_1 + h = 6.5 + 1.5 = 8$$
$$y_2 = y_1 + f(x_1, y_1)h = 1125 + f(6.5, 1125) \times 1.5$$
$$= 1125 + 6 \times 6.5^3 \times 1.5 = 3596.625 \approx y(8)$$

Hence,

$$\int_5^8 6x^3 dx = y(8) - y(5) \approx 3596.625 - 0 = 3596.625$$

c) i. Determine the *minimum* number of significant digits in each of these numerical values 3 + 2 (CO4) (PO1)

(a) 069420 (b) 6.94200×10^5 (c) 0.069420
(d) 69.00420 (e) 69.0 (f) 69420

Solution:
We need to determine the *minimum* number of significant digits.

(a) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.

(b) 6.94200×10^5 has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.

(c) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.

(d) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.

(e) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.

(f) 69420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.

ii. Determine the order and degree of each of these higher-order differential equations

(a) $x\dfrac{dy}{dx} - \left(\dfrac{d^2y}{dx^2}\right)^3 + 5 = 0$

(b) $\left(\dfrac{dy}{dx}\right)^2 - 4xy = 3$

(c) $\dfrac{d^3y}{dx^3} + x^4\dfrac{d^2y}{dx^2}\left(\dfrac{dy}{dx}\right)^4 = \sin x$

(d) $\dfrac{dy}{dx} + xy = x^2$

Solution:
The orders and degrees of these higher-order ODEs are,

(a) Order $= 2$, Degree $= 3$.

(b) Order $= 1$, Degree $= 2$.

(c) Order $= 3$, Degree $= 1$.

(d) Order $= 1$, Degree $= 1$.

d) i. Determine the LU Decomposition of the 3×3 matrix $5 + 5$ (CO2) (PO1)

$$A = \begin{bmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{bmatrix}$$

Solution:
The LU Decomposition can be done as follows,

$$\begin{bmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{bmatrix} \xrightarrow[E_{21}]{R_2 = -\frac{R_1}{3} + R_2} \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 4 & 15 & 3 \end{bmatrix} \xrightarrow[E_{31}]{R_3 = -\frac{2R_1}{3} + R_3} \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 0 & 3 & 1 \end{bmatrix} \xrightarrow[E_{32}]{R_3 = -\frac{R_2}{2} + R_3} \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where,

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{2}{3} & 0 & 1 \end{bmatrix} \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix}$$

So we have,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{2}{3} & \frac{1}{2} & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

ii. Suppose $u = u(x, y)$ is a function of two variables that we only know at discrete grid points (x_i, y_j) given in the matrix

$$[u_{i,j}] = \begin{bmatrix} 5.1 & 6.5 & 7.5 & 8.1 & 8.4 \\ 5.5 & 6.8 & 7.8 & 8.3 & 8.9 \\ 5.5 & 6.9 & 9.0 & 8.4 & 9.1 \\ 5.4 & 9.6 & 9.1 & 8.6 & 9.4 \end{bmatrix}$$

Find the approximate value of the following partial derivatives

(a) $u_x(x_2, y_4)$ (b) $u_y(x_2, y_4)$ (c) $u_{xx}(x_2, y_4)$

(d) $u_{yy}(x_2, y_4)$ (e) $u_{xy}(x_2, y_4)$

using the central difference formula. Consider $h = 0.5$ and $k = 0.2$.
Note: Assume 1-based indexing.

Solution:

(a) $u_x(x_2, y_4) \approx \dfrac{u_{3,4} - u_{1,4}}{2h} = \dfrac{8.4 - 8.1}{2 \times 0.5} = 0.3.$

(b) $u_y(x_2, y_4) \approx \dfrac{u_{2,5} - u_{2,3}}{2k} = \dfrac{8.9 - 7.8}{2 \times 0.2} = 2.75.$

(c) $u_{xx}(x_2, y_4) \approx \dfrac{u_{3,4} - 2u_{2,4} + u_{1,4}}{h^2} = \dfrac{8.4 - 2 \times 8.3 + 8.1}{0.5^2} = -0.4.$

(d) $u_{yy}(x_2, y_4) \approx \dfrac{u_{2,5} - 2u_{2,4} + u_{2,3}}{k^2} = \dfrac{8.9 - 2 \times 8.3 + 7.8}{0.2^2} = 2.5.$

(e) $u_{xy}(x_2, y_4) \approx \dfrac{u_{3,5} - u_{3,3} - u_{1,5} + u_{1,3}}{4hk} = \dfrac{9.1 - 9.0 - 8.4 + 7.5}{4 \times 0.5 \times 0.2} = -2.$